

Shrestha Dey

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SUMMARY

Computer Science undergraduate specializing in Artificial Intelligence & Machine Learning, with hands-on experience in deep learning, computer vision, and end-to-end AI application development. Proficient in Python, PyTorch, LLMs and RAG. Demonstrated ability to deliver end-to-end ML systems with measurable performance improvements. Committed to engineering intelligent, scalable AI solutions that generate real-world impact.

TECHNICAL SKILLS

Languages: Python, Java, SQL, JavaScript

AI / ML & Deep Learning: PyTorch, TensorFlow, Hugging Face Transformers, LangChain, OpenCV, RAG

Data & Visualization: NumPy, Pandas, Matplotlib

Web & API Development: React, FastAPI

Tools: Git, GitHub, Google Colab, VS Code

CS Fundamentals: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems (OS), Computer Networks, System Design

WORK EXPERIENCE

AI/ML Engineer Intern

October 2025 – January 2026

Yonder AI

Gurugram, India

- Architected and implemented **Liquid Neural Network (LNN)** models for time-dependent and dynamic student performance data, achieving real-time adaptive learning recommendations with measurably reduced prediction latency compared to traditional RNNs.
- Engineered end-to-end data pipelines encompassing ingestion, cleaning, feature engineering, and normalization, improving training data quality by **30%** and accelerating model convergence across multiple student cohort datasets.
- Spearheaded development of a personalized **student strength-weakness profiling model** using Liquid Neural Networks across 10+ academic subjects, enabling educators to deliver targeted interventions and improving student outcome prediction accuracy by **25%**.

PROJECTS

Kisan Sarthi – AI-Powered Agricultural Advisory Platform

March 2025 – May 2025

Machine Learning | NLP | RAG | End-to-End AI System

Python, FastAPI, FAISS, Streamlit

- Developed an AI advisory system using Retrieval-Augmented Generation (RAG) with FAISS-based vector search over an agricultural knowledge base for real-time query resolution.
- Built a modular FastAPI backend integrating chatbot, advisory engine, crop and pest data services, weather APIs, and multilingual translation for scalable decision support.
- Deployed and integrated a Telegram bot using webhooks with session management and message routing along with a Streamlit-based interface for interactive user access.

Crowd Detection and Management System

December 2025 – Present

Computer Vision | Full-Stack | IoT System

Python, PyTorch, YOLOv8, React, Node.js, MongoDB

- Deployed a real-time crowd density estimation system using YOLOv8, processing live camera feeds with sub-300ms latency for scalable public space monitoring.
- Built a full-stack dashboard with React and Node.js using WebSockets for real-time streaming, visualizing crowd density heatmaps and automated alerts for congestion control.
- Designed a crowd management module with dynamic routing signals to guide movement and reduce congestion in high-density zones.

EDUCATION

VIT Bhopal University

B.Tech in Computer Science – Specialization: Artificial Intelligence & Machine Learning | CGPA: 9.00/10

Bhopal, Madhya Pradesh

Sept 2023 – Sept 2027

Narbheram Hansraj English School

Higher Secondary Education (Class XII) | Aggregate: 92.7%

Jamshedpur, Jharkhand

March 2023

CERTIFICATIONS

- Model Context Protocol: Advanced Topics** – Anthropic
- Applied Machine Learning in Python** – Coursera (University of Michigan)
- AI Fluency for Students** – Anthropic